Potentiometer

DETECT | RASPBERRY PI | INPUT

What does it do?

A potentiometer lets your code read how far a knob has been turned.

This small version is called a trimpot. Turning the blue knob moves a contact called the wiper, changing the voltage your board reads.

Use it to choose a speed, a brightness or a sound. Turn gently and stop at the ends. The marking 103 means 10 kΩ.

Uses



Security-light adjustment

DETECT: Read sensitivity setting → DECIDE: Is movement strong enough? → DO: Switch the light on



Alien voice tuner

DETECT: Knob position → DECIDE: Should it rumble or squeak? → DO: Rumble or squeak



Knob	Outer pair	Wiper
Turn to adjust	Pins 1 and 3	Pin 2, offset

Code

Raspberry Pi

```
from gpiozero import MCP3008

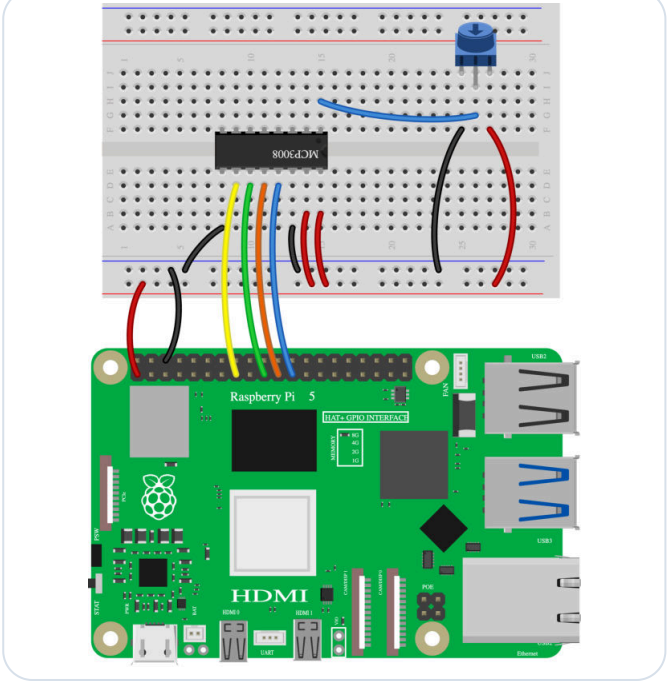
pot = MCP3008(0, select_pin=22)

setting = pot.value
```

`setting` runs from 0 to 1. Read `pot.value` again after turning; `pot.voltage` gives volts. The Pi needs the MCP3008 converter shown. Power the trimpot and converter from **3V3**.

Wiring

PI 5 + MCP3008



Wiper → **CH0**. CLK / DIN / DOUT / CS → **GPIO 11 / 10 / 9 / 22** (pins 23 / 19 / 21 / 15). **3V3** (pin 1), **GND** (pin 6). Chip notch faces right.